

2.5 The Precise Definition of a Limit

The definition of a limit presented in section 2.2 is vague...too vague. What do the following phrases mean?:

"x is close to 1"

"make the values of $f(x)$ arbitrarily close to L "

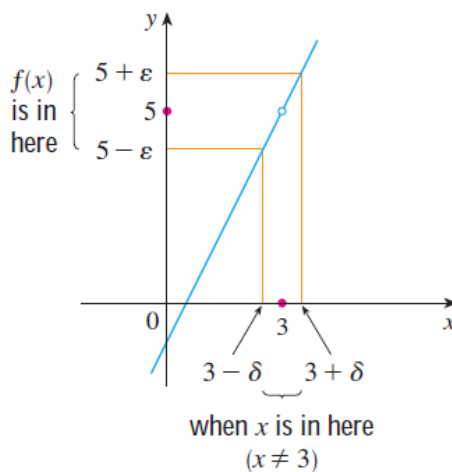
We wish to add some rigor and exactitude to these types of statements.

General Idea:

How could we show that $\lim_{x \rightarrow 3} (2x - 1) = 5$?

Notice that in the picture at the right we measure closeness using two different Greek letters:

- δ (delta) is used to measure
- ε (epsilon) is used to measure



We now develop the following rigorous definition.

Definition: Let f be a function defined on some open interval that contains the number a , except possibly a itself. Then we say that the limit of $f(x)$ as x approaches a is L , and we write $\lim_{x \rightarrow a} f(x) = L$ when the following holds:

Note: Determine the interval communicated by each of the following absolute value inequalities.

a) $|x - a| < \delta$

b) $|f(x) - L| < \varepsilon$

Using this knowledge we could rewrite the intervals in the previous definition in the following way.

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Note: Writing our intervals using absolute value inequalities will actually make it easier to verify the value of a limit.

Example 1: Consider the motivating example where we looked at $\lim_{x \rightarrow 3} (2x - 1) = 5$. Suppose that someone challenged you to show that the y -values of the function $f(x) = 2x - 1$ can get within 2 units from 5. Determine a value of δ that would make this possible.

Example 2: Prove that $\lim_{x \rightarrow 3} (2x - 1) = 5$ using only the definition of limit.

Example 3: Prove that $\lim_{x \rightarrow -2} (4x + 3) = -5$ using only the definition of limit.

Note: See section 2.5 of the textbook for more examples of an epsilon-delta proofs.

Note: There are similar rigorous epsilon delta definitions for one sided limits. We now present the rigorous definition of an infinite limit (we present ∞ , $-\infty$ is similar).

Definition: Let f be a function defined on some open interval that contains the number a , except possibly a itself. Then, we write $\lim_{x \rightarrow a} f(x) = \infty$ when the following holds:

For every number $M > 0$ there is a number $\delta > 0$ such that if $|x - a| < \delta$, then $f(x) > M$.

Note: If time permits, the instructor will explore the example of $\lim_{x \rightarrow 0} \ln\left(\frac{1}{x^2}\right) = \infty$ and show how it intuitively relates to the definition above.

Note: We can easily adjust the definition of above to define $\lim_{x \rightarrow a} f(x) = -\infty$ by requiring that $f(x) < M$.